Thursday Warm-up: Electrostatics

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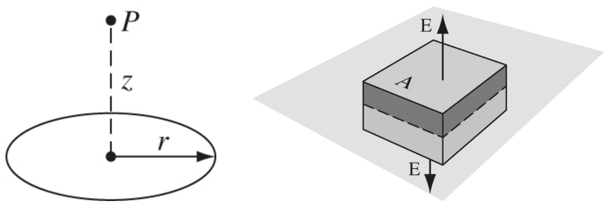

Figure 1: (Left) A ring of charge, with density λ. (Right) An infinite plane of charge, with density σ.

1 Memory Bank

- Definition of electric field: $\mathbf{F} = q\mathbf{E}$.
- Permittivity of free space: $\epsilon_0 = 8.85\times10^{-12}$ $\mathrm{N^{-1}C^2m^{-2}}$
- Let $\rho(\mathbf{r}')$ represent the charge density.
- Let $Q_{\text{enc}} = \int \rho d\tau'$, the total charge within a volume.
- Coulomb's Law, discrete

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{\mathscr{r}_i^2} \hat{\boldsymbol{\mathscr{r}}} \tag{1}$$

- Coulomb's Law, continuous

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{\mathscr{r}^2} \hat{\boldsymbol{\mathscr{r}}} \, d\tau' \tag{2}$$

- Gauss' Law, integral form

$$\oint \mathbf{E}) \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} \tag{3}$$

- Gauss' Law, differential form

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \tag{4}$$

2 Continuous Charge Distributions

1. Consider Fig. 1, (left). Find $\mathbf{E}$ at the point P, above the ring of charge that lies in the xy-plane, centered at the origin.
2. Consider again Fig. 1, (left). Suppose the portion of the ring with $y \geq 0$ was positive, and the portion with $y < 0$ was negative. What is the electric field?

3 Gauss' Law

1. Consider Fig. 1, (right). (a) Find $\mathbf{E}$ above the plane of positive charge, with charge density σ. (b) What is the field below the plane?
2. Imagine a *capacitor*, composed of a two parallel planes of charge, one positive and one negative. What is the field in the region between them? (b) If the planes are infinitely large, what is the field outside the capacitor?

4 Review

1. If a charge of q and mass m is placed within the capacitor, what is the acceleration of the charge?